

SSC CGL | 30-DAY PRELIMS REVISION PLAN

DAY 8

CONSOLIDATED PREP NOTES

Quant: Average & Alligation

Reasoning: Direction Sense

English: Idioms & Phrases

GK: Indian Geography

Formulas • Solved PYQs • Quick-Revision Tables • Exam Tricks

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How to use these notes: each section opens with core theory and formulas, followed by fully worked previous-year-style questions, and closes with a quick-revision table and exam-day tricks. Read theory once, then drill the PYQs against the clock.

QUANTITATIVE APTITUDE — SECTION 1

Average, Mixture & Alligation**1. Average — Basic Formula**

The average (arithmetic mean) of a set of numbers is the sum of all values divided by the count of values.

$$\text{Average} = \text{Sum of all quantities} \div \text{Number of quantities}$$

Key Properties

- The average of a set always lies between the smallest and the largest value.
- **Sum formula:** Sum = Average × Number of quantities.
- **Weighted average** is used when quantities carry different weights.

$$\text{Weighted Average} = (w_1x_1 + w_2x_2 + \dots + w_nx_n) \div (w_1 + w_2 + \dots + w_n)$$

Deviation Method (Shortcut)

A fast technique for SSC CGL: instead of computing the absolute sum, work with each value's deviation from an assumed average.

Worked Example. Deviation method for a changing average

The average score of three students A, B and C is 50. When a fourth student D joins, the average becomes 47. Find D's score.

New average = 47.

Each of the 3 original students is 3 units above the new average ($50 - 47 = 3$).

Total surplus to be balanced = $3 \times 3 = 9$.

D must be 9 units below the new average = $47 - 9 = 38$.

Answer: D scored 38

SSC CGL Type Question on Average**Q. Combining two groups with different averages**

The average age of 15 children is 10 years. When 5 more children with average age 8 years join them, what is the new average?

Total age of 15 children = $15 \times 10 = 150$.

Total age of 5 children = $5 \times 8 = 40$.

New total = 190; New count = 20.

New average = $190 \div 20 = 9.5$ years.

Answer: 9.5 years

2. Mixture & Alligation

Alligation is a rule for finding the ratio in which two or more ingredients, at given prices or concentrations, must be mixed to produce a mixture with a desired average price or concentration.

Alligation Rule (Cross Method)

If ingredients with values C_1 (cheaper) and C_2 (dearer) are mixed to give a mixture with mean value M :

$$\text{Quantity of Cheaper} : \text{Quantity of Dearer} = (C_2 - M) : (M - C_1)$$

Note: the mean value M always lies between C_1 and C_2 .

The “Beam Balance” Visualization

Picture a beam balance with equal arms and the mean value as the pivot. Distance from the mean is inversely proportional to quantity — if the distance ratio is 2:3, the quantity ratio is 3:2.

Important Points

- Mean value always lies between the two individual values.
- The closer the mean is to one value, the more of that component is used.
- Works for percentages, concentrations, prices, speeds, and similar quantities.
- Total value of the mixture equals the sum of the values of all ingredients.

3. Types of Alligation Problems

Type 1 — Simple Mixtures (Two Ingredients)

$$\text{Quantity of A} : \text{Quantity of B} = (C_B - M) : (M - C_A)$$

Type 2 — Complex Mixtures (More than Two Ingredients)

Apply alligation in steps. For concentrations a, b, c with mean m :

- Find the ratio for $a : b$ using alligation $\rightarrow x : y$
- Find the ratio for $b : c$ using alligation $\rightarrow p : q$
- Final ratio = $x : (y + p) : q$

Type 3 — Replacement Problems

Used when a fixed amount of mixture is repeatedly removed and replaced with another component.

$$\text{Final Quantity} = \text{Original Quantity} \times (1 - \text{Amount Removed} / \text{Total Quantity})^n$$

where n = number of times the process is repeated.

Worked Example. Repeated replacement

A vessel contains 80 L of ethanol. 20 L is removed and replaced with water. This is done twice. How much ethanol remains?

$$\text{Final ethanol} = 80 \times (1 - 20/80)^2 = 80 \times (3/4)^2 = 80 \times 9/16 = 45 \text{ L.}$$

$$\text{Water} = 80 - 45 = 35 \text{ L.}$$

Answer: 45 L ethanol remains (35 L water)

Type 4 — Mixture of Mixtures

Used when two mixtures, each containing the same two components in different ratios, are combined.

If mixture 1 has ratio $a:b$ and mixture 2 has ratio $x:y$, mixed in quantities $M:N$, then for ingredient A:

$$Q_A : Q_B = [M \cdot (a/(a+b)) + N \cdot (x/(x+y))] : [M \cdot (b/(a+b)) + N \cdot (y/(x+y))]$$

4. SSC CGL Previous Year Questions

Q1. Direct Alligation

In what ratio should tea costing ₹300/kg be mixed with tea costing ₹200/kg so that the mixture costs ₹225/kg?

$C_1 = 200, C_2 = 300, M = 225$.

Ratio = $(300 - 225) : (225 - 200) = 75 : 25 = 3 : 1$.

Answer: 3 : 1

Q2. Replacement Type

A drum contains 80 litres of ethanol. 20 litres is removed and replaced with water; this is repeated once more. How much water is present now?

Ethanol remaining = $80 \times (1 - 20/80)^2 = 80 \times 9/16 = 45$ L.

Water = $80 - 45 = 35$ L.

Answer: 35 L

Q3. Two Mixtures Mixed

Two equal vessels are filled with a mixture of water and milk in ratios 3:4 and 5:3 respectively. If poured into a third vessel, find the new ratio of water to milk.

Vessel 1: Water = $3/7$, Milk = $4/7$. Vessel 2: Water = $5/8$, Milk = $3/8$.

Equal quantities $\rightarrow$ ratio = $(3/7 + 5/8) : (4/7 + 3/8) = (24+35)/56 : (32+21)/56$.

Answer: 59 : 53

Q4. Dishonest Milkman

A dishonest milkman mixes water with milk costing ₹35/L to make 30% profit, selling the mixture at ₹39/L. Find the ratio of water to milk.

Cost price of mixture = $SP \div (1 + \text{profit}\%) = 39 \div 1.3 = ₹30/L$.

C_1 (water) = 0, C_2 (milk) = 35, $M = 30$.

Ratio = $(35 - 30) : (30 - 0) = 5 : 30 = 1 : 6$.

Answer: 1 : 6

Q5. Withdrawal and Replacement

A mixture contains milk and water in ratio 3:1. 20 L of the mixture is replaced by water, and the new ratio becomes 1:1. Find the initial quantity.

Let initial quantity = $4x$ (milk = $3x$, water = x).

Removed milk = $20 \times 3/4 = 15$ L; removed water = 5 L.

New milk = $3x - 15$; new water = $x - 5 + 20 = x + 15$.

$(3x - 15) : (x + 15) = 1 : 1 \rightarrow 3x - 15 = x + 15 \rightarrow 2x = 30 \rightarrow x = 15$.

Answer: Initial quantity = $4x = 60$ L

Q6. Acid–Water Mixture

The ratio of acid to water in a mixture is 1:3. If 5 L acid is added, the new ratio becomes 1:2. Find the quantity of the new mixture.

Let acid = x , water = $3x$. After adding 5 L: $(x + 5) : 3x = 1 : 2$.

$2x + 10 = 3x \rightarrow x = 10$.

New mixture = $(10 + 5) + 30 = 45$ L.

Answer: 45 L

5. Formula Summary

Concept	Formula
Average	$\text{Sum} \div \text{Count}$
Weighted Average	$\Sigma(w \times x) \div \Sigma w$
Alligation Rule	$Q_1/Q_2 = (C_2 - M)/(M - C_1)$
Replacement (n times)	$\text{Final} = \text{Original} \times (1 - r/T)^n$
Mixture of Mixtures	Weighted average of component fractions

6. Quick Tips for SSC CGL

- ◆ The alligation diagram saves time — practice drawing it mentally.
- ◆ In replacement problems, always track the component that is NOT being added.
- ◆ For profit–loss mixtures, first find the CP of the mixture using SP and profit%.
- ◆ Roughly 1–2 questions from this topic appear in SSC CGL Prelims.
- ◆ Use the deviation method for averages to avoid long calculations.
- ◆ When three mixtures are involved, use the LCM method for equal-capacity bottles.

GENERAL INTELLIGENCE & REASONING — SECTION 2

Direction Sense Test**1. Basic Concepts — The Compass Directions**

There are 8 main directions, split into 4 cardinal and 4 ordinal (intermediate) directions.

4 Cardinal Directions

- **North (N)** — upward
- **South (S)** — downward
- **East (E)** — rightward
- **West (W)** — leftward

4 Ordinal (Intermediate) Directions

- **North-East (NE)** — between N and E (45°)
- **North-West (NW)** — between N and W (45°)
- **South-East (SE)** — between S and E (45°)
- **South-West (SW)** — between S and W (45°)

2. Angle Between Directions

Directions	Angle
Adjacent cardinal	90°
Opposite cardinal	180°
Cardinal to ordinal	45°
Opposite ordinal	180°
Perpendicular ordinal (e.g. NE to SE)	90°

3. Turning Concepts

Left turn = -90° (anticlockwise) • Right turn = $+90^\circ$ (clockwise) • 180° turn = opposite direction

From → Turn	Result
North → Left	West
East → Left	North
South → Left	East
West → Left	South
North → Right	East
East → Right	South
South → Right	West
West → Right	North

180° opposite pairs: North ↔ South, East ↔ West, NE ↔ SW, NW ↔ SE.

4. Shadow & Sun Concepts

Time	Sun Direction	Shadow Falls	If Facing North
Morning (sunrise)	East	West	Shadow on the left
Evening (sunset)	West	East	Shadow on the right
Noon (India)	Roughly South, overhead	No / very small shadow	—

5. Shorthand Notation

Always convert word problems into a simple notation before solving. Example: "Ramesh walks 10 km North, then turns right and walks 5 km" becomes: N → 10, then Right → E → 5.

6. Types of Direction Sense Questions

- **Type 1 — Final Direction Only:** track only the turns, ignore distances. Odd number of turns changes direction; even number may keep it same or opposite depending on angles.
- **Type 2 — Final Position / Displacement:** use coordinate geometry. Start at (0,0); North = +Y, South = -Y, East = +X, West = -X.
- **Type 3 — Minimum Distance (Shortest Path):** Minimum Distance = $\sqrt{x^2 + y^2}$, where x = net East–West displacement and y = net North–South displacement.
- **Type 4 — Direction of One Person Relative to Another:** find both positions, then the vector from A to B, and read its direction.
- **Type 5 — Shadow Based Questions:** morning → sun in East, shadow West; evening → sun in West, shadow East; noon → sun overhead, no shadow.
- **Type 6 — Clockwise / Anticlockwise from a Given Direction:** clockwise = right turns, anticlockwise = left turns; use angle addition/subtraction.

7. SSC CGL Previous Year Questions

Q1. Basic Path

A man walks 5 km East, turns right and walks 3 km, then turns right again and walks 5 km. How far is he from the starting point?

Start (0,0) → 5 km East → (5,0).

Right turn (South), 3 km → (5,-3).

Right turn (West), 5 km → (0,-3).

Distance from start = 3 km (South of start).

Answer: 3 km

Q2. Multiple Turns

Ravi walks 10 km North, turns left and walks 8 km, then turns left again and walks 5 km. In which direction is he from his house now?

Start (0,0) → 10 km North → (0,10).

Left turn (West), 8 km → (-8,10).

Left turn (South), 5 km → (-8,5).

Final position (-8,5) → X<0, Y>0 → North-West.

Answer: North-West

Q3. Minimum Distance

A person goes 30 m North, then 20 m East, then 30 m South. What is the shortest distance from the starting point?

Net North–South = 30 – 30 = 0 m. Net East–West = 20 m.

Distance = $\sqrt{(0^2 + 20^2)} = 20$ m.

Answer: 20 m

Q4. Shadow Based

In the morning, a man walks towards East. In which direction is his shadow?

Morning: sun is in the East; man faces East.

Shadow falls opposite to the sun, so it is behind him — to the West.

Answer: West (behind him)

Q5. Complex Path with Shadow

At 7:00 AM, a man starts walking and his shadow is to his right. In which direction is he walking?

7:00 AM = morning → sun in East → shadow falls in West.

Shadow is to his right → his right side corresponds to West → he is facing North.

Answer: North

Q6. Person Relative to Another

A is 10 m North of B. C is 6 m East of A. D is 8 m South of C. E is 12 m West of D. In which direction is E from B?

Let B = (0,0). Then A = (0,10), C = (6,10), D = (6,2), E = (-6,2).

From B(0,0) to E(-6,2): X = -6 (West), Y = 2 (North) → North-West.

Answer: North-West

Q7. Clockwise Rotation of Direction Names

If North is called East, East is called South, South is called West, and West is called North, then what is North-East called?

Mapping: $N \rightarrow E$, $E \rightarrow S$, $S \rightarrow W$, $W \rightarrow N$.

North-East lies between N and E; under the mapping this maps between E and S, i.e. South-East.

Answer: South-East

Q8. 45° Turn Problem

A person starts walking North-East. He turns 45° clockwise, then 90° anticlockwise, then 135° clockwise. In which direction is he facing now?

Start: NE (45° from North).

+45° clockwise $\rightarrow$ 90° from North $\rightarrow$ East.

-90° anticlockwise $\rightarrow$ 0° from North $\rightarrow$ North.

+135° clockwise $\rightarrow$ 135° from North $\rightarrow$ South-East.

Answer: South-East

Q9. Both Start from Same Point

A and B start from the same point. A walks 12 m East then 5 m North. B walks 12 m West then 9 m South. What is the distance between A and B?

A: $(0,0) \rightarrow (12,0) \rightarrow (12,5)$. B: $(0,0) \rightarrow (-12,0) \rightarrow (-12,-9)$.

Distance = $\sqrt{[(12-(-12))]^2 + (5-(-9))^2} = \sqrt{[24^2 + 14^2]} = \sqrt{[576+196]} = \sqrt{772} = 2\sqrt{193}$ m.

Answer: $2\sqrt{193}$ m

Q10. Through a Building

A man goes 15 m West, then 20 m North, then 8 m East. What is the minimum distance between the start and end points?

Start $(0,0) \rightarrow$ 15 m West $\rightarrow (-15,0) \rightarrow$ 20 m North $\rightarrow (-15,20) \rightarrow$ 8 m East $\rightarrow (-7,20)$.

Distance = $\sqrt{[(-7)^2 + 20^2]} = \sqrt{[49+400]} = \sqrt{449}$ m.

Answer: $\sqrt{449}$ m

8. Important Tricks for SSC CGL

- **Last Direction Shortcut:** count right and left turns, net turns = $|R - L|$, each turn = 90° ; add/subtract from the starting direction.
- **Coordinate Method (best for displacement):** track X (East +, West –) and Y (North +, South –) separately.
- **Shadow Memory Aid:** Morning → Sun East → Shadow West. Evening → Sun West → Shadow East. If facing North: morning shadow on the left, evening shadow on the right.
- **Right vs Left Confusion:** write “R” or “L” after every turn while solving.
- **Distance Calculation:** “How far” / “minimum distance” → use Pythagoras theorem; “in which direction” → use coordinate quadrant analysis.

9. Quadrant Rules

Coordinates	Direction
$X > 0, Y > 0$	North-East
$X < 0, Y > 0$	North-West
$X > 0, Y < 0$	South-East
$X < 0, Y < 0$	South-West
$X = 0, Y > 0$	North
$X = 0, Y < 0$	South
$X > 0, Y = 0$	East
$X < 0, Y = 0$	West

10. Common Mistakes to Avoid

- Confusing left/right while facing different directions.
- Skipping a rough diagram for complex, multi-turn questions.
- Ignoring the starting direction in 45° -turn problems.
- Forgetting that the sun's direction — and hence the shadow — changes with time of day.
- Using simple addition instead of Pythagoras for displacement questions.

11. Practice Strategy

- ◆ Attempt 5–7 questions daily to build speed.
- ◆ Draw diagrams for every complex question.
- ◆ Time yourself — aim for ≤ 30 seconds per question.
- ◆ Master the coordinate method — it is foolproof.
- ◆ Memorize the shadow rules — they save valuable time.
- ◆ Use answer options to verify quickly when stuck.

ENGLISH LANGUAGE — SECTION 3

Vocabulary: Idioms & Phrases

1. What Are Idioms & Phrases?

An idiom is a group of words whose meaning is figurative rather than literal — the expression means something different from its individual words. Example: “A blessing in disguise” does not mean a literal hidden blessing; it means a situation that seems bad at first but turns out well.

Idioms vs Phrases

Feature	Idioms	Phrases
Meaning	Figurative (cannot be understood literally)	Can be literal or figurative
Fixedness	Generally fixed expressions	More flexible, can be modified
Example	“Kick the bucket” (to die)	“Actions speak louder than words”

2. Why This Topic Matters for SSC CGL

- **Frequently asked:** typically 1–2 questions appear in the Tier-I English section.
- **Negative marking:** each wrong answer carries –0.50 marks.
- **High repeats:** many idioms recur across years, making preparation highly rewarding.
- **Context matters:** understanding the full meaning is key to picking the correct option.

3. Types of Idiom/Phrase Questions

- **Type 1 — Direct Meaning:** given an idiom/phrase, choose its correct meaning.
- **Type 2 — Context-Based:** given a sentence with the idiom/phrase in bold, choose its meaning as used in that sentence.

4. Important Idioms & Phrases

Frequently Asked (Previous Year Questions)

Idiom / Phrase	Meaning	Usage
Axe to grind	To have a hidden / personal agenda	"He has an axe to grind in this matter."
Keep at bay	To control something and prevent it causing problems	"The army kept the enemy at bay."
Put one's foot down	To be firm and inflexible in a decision	"Sarita put her foot down on the marriage issue."
Draw a blank	To fail to find or remember something	"The police drew a blank in their investigation."
Make a killing	To make a lot of money quickly	"He made a killing in the stock market."
Cut above	To be superior to someone/something	"Sachin's skills are a cut above the rest."
Keep in the loop	To keep someone regularly informed	"Keep me in the loop about the outcome."
Cut and dried	Ready-made; predetermined	"There is no cut and dried method."
Black out	To lose consciousness	"The man blacked out during the parade."
Hold your horses	To wait; be patient	"Hold your horses for a moment."

Commonly Tested Idioms

Idiom	Meaning
Beat around the bush	To avoid talking about what's important
On cloud nine	To be extremely happy
Under the weather	To feel sick
Bite the bullet	To accept something unpleasant and get it over with
Break the ice	To start a conversation in a social setting
Let the cat out of the bag	To reveal a secret
A blessing in disguise	A good thing that seemed bad at first
In the nick of time	Just in time
Add fuel to the fire	To make a situation worse
Sit on the fence	To remain neutral or undecided
Once in a blue moon	Very rarely
Kill two birds with one stone	To achieve two goals with one effort
Burn the midnight oil	To work or study late into the night
Take it with a grain of salt	To not take something too seriously
Face the music	To accept the consequences

Less Common but Important

Idiom / Phrase	Meaning
Gainsaying	To deny or declare something false
Make a fuss about	To show excessive attention/activity over something
Go off the air	To stop broadcasting a TV/radio program
Go to somebody's head	To make someone dizzy/drunk, or conceited
Make amends	To compensate for a mistake or wrong
High and dry	Left in a difficult situation without help
Grease the palm	To bribe someone
Neck and neck	Competitors very close in a contest
Salad days	Youthful period of enthusiasm and inexperience

5. SSC CGL Previous Year Questions

Q#	Idiom / Phrase in Context	Correct Answer
Q1	Get the message	Understand what is implied by a remark or action
Q2	Keep at bay	To control something and prevent it from causing you problems
Q3	"...always has an axe to grind"	has a private agenda
Q4	"...but drew a blank"	did not find him
Q5	"...Sarita put her foot down"	was firm
Q6	"...make him a cut above the rest"	superior to
Q7	"...make a killing in the stock market"	make money quickly
Q8	"...kindly keep me in the loop"	informed regularly
Q9	"No gainsaying the fact..."	denying
Q10	"No cut and dried method..."	ready made
Q11	"...blacked out during the parade"	lost consciousness
Q12	"Hold your horses..."	Wait

All items above are drawn from actual SSC CGL question patterns; each was tested with four options and the idiom's core figurative meaning was the correct choice.

6. Smart Preparation Strategies

- **Flashcards:** create physical or digital flashcards with the idiom, meaning and an example sentence for quick, frequent revision.
- **Group by theme:** organize idioms by meaning category — success, failure, happiness, anger, etc.
- **Use in sentences:** practice writing your own sentences to get comfortable with usage.
- **Real-life exposure:** notice idioms in social media, news articles and everyday conversation.
- **Learn opposites:** contrasting idioms reinforce memory for both.

Study Plan

- Daily practice: learn 5–10 idioms per day.
- Weekly revision: revisit every idiom learned that week.
- Mock tests: practice with previous year questions regularly.
- Focus area: prioritize idioms that have appeared in past SSC papers.

7. Common Mistakes to Avoid

- Guessing without context — always read the full sentence first.
- Confusing similar idioms, e.g. “Spill the beans” ≠ “Beat around the bush”.
- Word-to-word translation — never translate idioms literally.
- Not revising regularly — idioms need constant repetition for retention.
- Ignoring negative marking — use elimination strategy when unsure.

8. Quick Reference: Idiom Categories

Category	Idiom — Meaning
Success / Excellence	Cut above — superior to Make a killing — make a lot of money On the ball — doing a good job
Failure / Difficulty	Draw a blank — fail to find/remember Back to the drawing board — start over Throw in the towel — give up
Secrecy / Revelation	Let the cat out of the bag — reveal a secret Spill the beans — disclose a secret Zip your lip — keep quiet
Patience / Waiting	Hold your horses — wait/be patient Rome wasn't built in a day — big tasks take time
Action / Decision	Bite the bullet — accept something unpleasant Put one's foot down — be firm Sit on the fence — remain undecided

GENERAL AWARENESS — SECTION 4

Static GK: Indian Geography – States, Resources & Crops

1. Introduction & Weightage

Indian Geography is a highly scoring section in SSC CGL. Recent exam analysis shows Geography contributes 8–12 questions across Tier-I and Tier-II, with roughly 3–4 questions in Tier-I specifically on states, resources and crops. Questions are factual and often repeated, making strategic preparation highly rewarding.

2. Physiographic Divisions of India

Division	Key Features
The Himalayas	Young fold mountains; includes Karakoram (K2 – India's highest peak, 8611 m), Zaskar, Ladakh, Pir Panjal ranges
Northern Plains	Formed by the Ganga, Brahmaputra and Indus systems; fertile alluvial soil
Peninsular Plateau	Deccan Plateau (oldest landmass); Chhota Nagpur Plateau – rich in minerals
Western Coastal Plains	Narrow; includes Konkan, Kannad, Malabar coasts
Eastern Coastal Plains	Wider; includes Coromandel Coast; fertile deltas
Islands	Andaman & Nicobar (Bay of Bengal), Lakshadweep (Arabian Sea)

SSC CGL fact: the Deccan Plateau is the largest plateau in India. The Netarhat Mountains in Jharkhand are called the “Queen of Chotanagpur”.

3. Major River Systems & Tributaries

Himalayan Rivers

River	Origin	States / Regions	Key Tributaries
Ganga	Gangotri (Uttarakhand)	UP, Bihar, West Bengal	Yamuna, Ghaghara, Gandak, Kosi
Yamuna	Yamunotri (Uttarakhand)	Delhi, UP	Chambal, Betwa, Ken
Brahmaputra	Tibet (Yarlung Tsangpo)	Assam, Arunachal	Manas, Subansiri, Dhansiri
Indus	Tibet	J&K, Ladakh	Jhelum, Chenab, Ravi, Beas, Sutlej

Key fact: the Ganga is India's longest river at 2525 km. The Brahmaputra carries the highest volume among Indian rivers and is known as the Yarlung Tsangpo in Tibet.

PYQ 2017. River identification

The Brahmaputra is also called: (a) Burlun-buthur (b) Tasangpo-Brahmaputra (c) Yarlung–Tsangpo (d) All of the above

Answer: (d) All of the above

Peninsular Rivers

River	Origin	States	Key Facts
Godavari	Nasik (Maharashtra)	Maharashtra, Telangana, AP	Largest peninsular river; "Dakshina Ganga"
Krishna	Mahabaleshwar (Maharashtra)	Maharashtra, Karnataka, AP, Telangana	Nagarjuna Sagar Project built on this river
Kaveri	Talakaveri (Karnataka)	Karnataka, Tamil Nadu	"Ganga of the South"
Mahanadi	Chhattisgarh	Chhattisgarh, Odisha	Hirakud Dam

PYQ 2017. Dam identification

The Nagarjuna Sagar project is constructed on which river? (a) Kaveri (b) Brahmaputra (c) Krishna (d) Ganga

Answer: (c) Krishna

4. Mineral Resources & Distribution

Mineral	Leading States	Notes
Iron Ore (Haematite)	Odisha (largest), Jharkhand, Chhattisgarh, Karnataka, Goa	Most common iron ore; Odisha has the highest reserves
Coal	Jharkhand, Odisha, Chhattisgarh	—
Manganese	Odisha (Sundargarh), Maharashtra, Karnataka	—
Bauxite	Odisha, Gujarat, Jharkhand	—
Mica	Jharkhand, Rajasthan, Andhra Pradesh	—
Petroleum	Digboi (Assam), Mumbai High, Gujarat	Digboi is the oldest oil field in India
Monazite (Thorium)	Kerala (Palakkad and Kollam)	World's richest deposits; a nuclear mineral
Gold	Kolar (Karnataka), Hutti (Karnataka)	Hutti is another important gold mine

PYQ. Highest haematite reserves

Which state has the highest reserves of haematite ore? (a) Rajasthan (b) Tamil Nadu (c) Maharashtra (d) Odisha

Answer: (d) Odisha

Mineral Belts of India

The Chhota Nagpur Belt (Jharkhand, Odisha, Chhattisgarh, West Bengal, Bihar) is rich in iron ore, coal, mica, bauxite and copper, and is often called the “Ruhr of India”.

PYQ. Chhota Nagpur belt minerals

Which mineral is NOT found in the Chhota Nagpur belt? (a) Gold (b) Iron-ore (c) Chromite (d) Coal

Answer: (a) Gold

5. Agriculture & Crop Patterns

Crop Seasons

Season	Crops	Sowing	Harvesting
Kharif	Rice, Cotton, Sugarcane, Groundnut	June–July (Monsoon)	Sept–Oct
Rabi	Wheat, Barley, Mustard, Gram	Oct–Nov (Winter)	March–April
Zaid	Watermelon, Muskmelon, Fodder	Summer	—

Food Crops — Major Producing States

Crop	Leading States
Rice	West Bengal, UP, Punjab, Tamil Nadu
Wheat	UP, Punjab, Haryana, MP
Jowar	Maharashtra, Karnataka, MP
Bajra	Rajasthan, Gujarat, UP
Maize	Bihar, MP, Karnataka, Rajasthan

Pulses, Oilseeds & Fibre Crops

Crop	Leading States	Notes
Gram (Chana)	MP, UP, Maharashtra, Rajasthan	Most important pulse crop
Tur (Arhar)	Maharashtra (1/3 output), UP, Karnataka, Gujarat, MP	Second most important pulse
Groundnut	Gujarat, Rajasthan, TN, Telangana, AP, Karnataka, Maharashtra	Rainfed Kharif crop
Rapeseed / Mustard	Rajasthan (1/3), Haryana, MP	Rabi season crop
Soyabean	MP & Maharashtra	~90% of national production
Sunflower	Karnataka, AP, Telangana, Maharashtra	—
Cotton	Gujarat, Maharashtra, Telangana	India ranks 2nd worldwide after China; needs clear sky while flowering
Jute	West Bengal (3/4 output), Bihar, Assam	India produces ~3/5 of world jute

Commercial Crops

Crop	Leading States	Notes
Sugarcane	UP (2/5 output), Maharashtra, Karnataka, Tamil Nadu, AP	India ranks 2nd worldwide after Brazil
Tea	Assam, West Bengal	—
Coffee	Karnataka, Kerala, Tamil Nadu	—
Rubber	Kerala	Leading rubber-producing state

Soil Types & Crops

Soil Type	Regions	Suited Crops
Alluvial	Northern Plains	Rice, Wheat, Sugarcane
Black (Regur)	Maharashtra, Gujarat, MP, TN, AP, Karnataka	Cotton (best suited)
Laterite	West Bengal, Kerala, Tamil Nadu	Tea, Coffee, Rubber — plantation crops
Red Soil	Tamil Nadu, Karnataka, Andhra Pradesh	—

PYQ. Black soil distribution

Black soil is NOT found in: (a) Tamil Nadu (b) Maharashtra (c) West Bengal (d) Andhra Pradesh

Answer: (c) West Bengal

6. National Parks & Wildlife

Park / Sanctuary	State	Key Animal
Sundarbans	West Bengal	Royal Bengal Tiger; world's largest mangrove forest
Bandhavgarh	Madhya Pradesh	Tiger
Jaldapara	West Bengal	Indian one-horned Rhinoceros
Kaziranga	Assam	One-horned Rhinoceros
Periyar	Kerala	Tiger, Elephant

PYQ 2017. Largest mangrove forest

Largest mangrove forest in the world? (a) Pichavaram (b) Pitara (c) Sundarbans (d) Bhitarkanika

Answer: (c) Sundarbans

7. Important Lakes & Ports

Major Lakes

Lake	State	Type
Wular Lake	Jammu & Kashmir	Largest freshwater lake in India
Vembanad Lake	Kerala	Longest lake in India
Chilika Lake	Odisha	Largest brackish-water lake

Major Ports

India has 12 major ports, including:

- **Mumbai (Maharashtra)** — natural harbor; India's largest port.
- **Kochi (Kerala)** — known as the “Queen of the Arabian Sea”.
- **Chennai (Tamil Nadu)** — an artificial port.
- **Visakhapatnam (Andhra Pradesh)** — the deepest port.
- **Kolkata (West Bengal)** — an inland riverine port.
- Others include Paradip (Odisha) and Marmugao (Goa).

PYQ. Queen of the Arabian Sea

Which is called “The Queen of the Arabian Sea”? (a) Mumbai (b) Thiruvananthapuram (c) Kochi (d) Surat

Answer: (c) Kochi

8. Key Quick Facts

Feature	Fact
Longest river in India	Ganga (2525 km)
Highest peak in India	K2 (Godwin Austen) – 8611 m
Largest state (area)	Rajasthan
Largest state (population)	Uttar Pradesh
Smallest state (area)	Goa
State with most coastline	Gujarat
Largest delta	Sundarbans Delta
Largest freshwater lake	Wular Lake (J&K;)
Oldest oil field	Digboi (Assam)

9. Quick Revision Tips

- ◆ Map practice: mark rivers, mineral belts and crop regions on a blank map of India.
- ◆ Remember soil–crop correlations, e.g. Black soil → Cotton; Laterite → plantation crops.
- ◆ Use the tables in this section for rapid recall before the exam.
- ◆ Drill previous year questions — patterns repeat frequently.

“Indian Geography is highly factual and repetitive in SSC CGL. Focus on producer states for crops and minerals, soil–crop correlation, and river locations for maximum marks.”